# PROJECT REPORT

Smart Parking Allocation System

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**Course** – Fundamentals of AI & ML

**Course Code** – CSA2001

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# 1. Introduction :

My aim behind this project is to solve a daily life problem of most of the car owners.

The Smart Parking Allocation System is a Python based project designed to efficiently manage parking spaces. In many urban areas, finding a parking slot is time-consuming and frustrating . This system helps in identifying the nearest available parking slot automatically.

The project uses the concept of pathfinding and data structures to guide vehicles from the entry point to the nearest empty slot. It improves time efficiency and reduces congestion in parking areas.

# 2. Objective:

- To develop a system that finds the nearest parking slot .
- To reduce time spent searching for parking .
- To implement Breadth First Search (BFS) algorithm.
- To display movement path for the vehicle.
- To calculate parking charges based on time .

# 3. Methodology:

The System uses the Breadth First Search (BFS) algorithm to find the shortest path from the entry point to the nearest available parking slot .

- The parking area is represented as a grid (matrix)
- Each cell can be:
  - E → Entry point
  - 0 → Empty slot
  - X → Blocked space
- BFS explores all possible directions:
  - Up
  - Down
  - Left
  - Right
- It ensures the shortest path is found first

# 4. Technologies Used:

- Programming Language: Python

- Concept Used: Data Structures (Queue using deque)
- Algorithm: Breadth First Search (BFS)

# 5. System Design:

- **Input:**
  - Serial  Number
  - Car Number
  - Owner Number
  - Parking Hours
- **Process**:
  - Calculate parking charges
  - Identify entry point
  - Apply BFS to find nearest slot
  - Track movement path
- **Output**:
  - Nearest parking slot location
  - Step by Step movement path
  - Total parking charges

# 6. Algorithm:

1.Start

2.Input user details

3.Initialize parking grid

4.Find entry point (E)

5.Apply BFS:

- Use queue to explore nodes
- Mark visited cells

6.If empty slot found :

- Print location
- Print path

7.Else:

- Display no slot available

8.End.

# 7. Implementation (Code Summary):

The program is implemented using Python. A queue is used to perform BFS traversal. The system tracks visited nodes and stores the movement path. When an empty slot is found , the program prints the shortest path along with directions.

# 8. Output:

```
PS C:\Users\ABOLI GUNJAL\Vs Code all> & "C:/Users/ABOLI GUNJAL/AppData/Local/Programs/Python/Python313/python.exe" "c
:/Users/ABOLI GUNJAL/Vs Code all/main.py/Smart parking Allocation System.py"
===== Smart Parking Allocation System =====
Details:
Enter serial_no:2
car_num:MH15AB4746
owner_num:9365652552
park_hrs:4
Charges: 120

Nearest Slot Found at: (0,1)
Car Movement Path:
RIGHT → PARKED 🚗
PS C:\Users\ABOLI GUNJAL\Vs Code all>
```

# 9. Conclusion:

The Smart Parking Allocation System successfully demonstrates the use of BFS algorithm in solving real-world problems. It efficiently finds the nearest parking slot and provides a clear path for navigation. This project helps in understanding practical applications of data structures and algorithms.